# QUIMPER BRETAGNE AP, France

WMO: **072010**

Lat: **47.9728N** Lon: **4.1606W** Elev: **91** StdP: **100.24** Time Zone: **1.00 (EUC)** Period: **99-19** WBAN: **99999**

## Annual Heating, Humidification, and Ventilation Design Conditions

| | Coldest Month | Heating DB | | Humidification DP/MCDB and HR | | | | | | Coldest Month WS/MCDB | | | | MCWS/PCWD to 99.6% DB | | WSF | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | | | 99.6% | | | 99% | | | 0.4% | | 1% | | | | | |
| | | 99.6% | 99% | DP | HR | MCDB | DP | HR | MCDB | WS | MCDB | WS | MCDB | MCWS | PCWD | | |
| | (a) | (b) | (c) | (d) | (e) | (f) | (g) | (h) | (i) | (j) | (k) | (l) | (m) | (n) | (o) | (p) | |
| (1) | **2** | **-1.4** | **-0.1** | **-5.4** | **2.4** | **0.5** | **-3.7** | **2.8** | **1.5** | **13.3** | **9.6** | **12.2** | **9.9** | **2.9** | **40** | **0.561** | (1) |

## Annual Cooling, Dehumidification, and Enthalpy Design Conditions

| | Hottest Month | Hottest Month DB Range | Cooling DB/MCWB | | | | | | Evaporation WB/MCDB | | | | | | MCWS/PCWD to 0.4% DB | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | | 0.4% | | 1% | | 2% | | 0.4% | | 1% | | 2% | | | | |
| | | | DB | MCWB | DB | MCWB | DB | MCWB | WB | MCDB | WB | MCDB | WB | MCDB | MCWS | PCWD | |
| | (a) | (b) | (c) | (d) | (e) | (f) | (g) | (h) | (i) | (j) | (k) | (l) | (m) | (n) | (o) | (p) | |
| (2) | **7** | **8.0** | **27.0** | **19.4** | **25.0** | **18.3** | **23.1** | **17.5** | **20.2** | **25.3** | **19.2** | **23.3** | **18.3** | **21.6** | **3.6** | **50** | (2) |

| | Dehumidification DP/MCDB and HR | | | | | | | | | Enthalpy/MCDB | | | | | | Extreme Max WB | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | 0.4% | | | 1% | | | 2% | | | 0.4% | | 1% | | 2% | | | |
| | DP | HR | MCDB | DP | HR | MCDB | DP | HR | MCDB | Enth | MCDB | Enth | MCDB | Enth | MCDB | | |
| | (a) | (b) | (c) | (d) | (e) | (f) | (g) | (h) | (i) | (j) | (k) | (l) | (m) | (n) | (o) | (p) | |
| (3) | **18.6** | **13.6** | **21.6** | **17.8** | **12.9** | **20.4** | **17.1** | **12.4** | **19.6** | **58.2** | **25.7** | **54.8** | **23.4** | **52.1** | **21.5** | **24.9** | (3) |

## Extreme Annual Design Conditions

| | Extreme Annual WS | | | | Extreme Annual Temperature | | | | n-Year Return Period Values of Extreme Temperature | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | | | | Mean | | Standard Deviation | | n=5 years | | n=10 years | | n=20 years | | n=50 years | | |
| | 1% | 2.5% | 5% | | Min | Max | Min | Max | Min | Max | Min | Max | Min | Max | Min | Max | |
| | (a) | (b) | (c) | | (e) | (f) | (g) | (h) | (i) | (j) | (k) | (l) | (m) | (n) | (o) | (p) | |
| (4) | **11.0** | **9.6** | **8.6** | DB | **-3.7** | **30.0** | **1.8** | **2.4** | **-5.0** | **31.8** | **-6.0** | **33.2** | **-7.0** | **34.5** | **-8.3** | **36.3** | (4) |
| (5) | | | | WB | **-4.6** | **22.2** | **1.8** | **1.3** | **-5.9** | **23.2** | **-6.9** | **24.0** | **-7.9** | **24.7** | **-9.2** | **25.7** | (5) |

## Monthly Climatic Design Conditions

| | | | | Annual | Jan | Feb | Mar | Apr | May | Jun | Jul | Aug | Sep | Oct | Nov | Dec | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | | | (d) | (e) | (f) | (g) | (h) | (i) | (j) | (k) | (l) | (m) | (n) | (o) | (p) | |
| (6) | Temperatures, Degree-Days and Degree-Hours | | DBAvg | **11.9** | **7.0** | **7.0** | **8.6** | **10.4** | **13.0** | **15.9** | **17.5** | **17.2** | **15.7** | **13.0** | **9.7** | **7.7** | (6) |
| (7) | | | DBStd | **4.66** | **2.95** | **2.84** | **2.65** | **2.68** | **2.66** | **2.64** | **2.41** | **2.11** | **2.31** | **2.50** | **2.57** | **3.11** | (7) |
| (8) | | | HDD10.0 | **399** | **99** | **88** | **58** | **25** | **4** | **0** | **0** | **0** | **0** | **5** | **36** | **83** | (8) |
| (9) | | | HDD18.3 | **2391** | **353** | **318** | **301** | **238** | **166** | **82** | **46** | **47** | **84** | **166** | **261** | **330** | (9) |
| (10) | | | CDD10.0 | **1099** | **5** | **4** | **15** | **38** | **98** | **178** | **232** | **224** | **172** | **97** | **26** | **12** | (10) |
| (11) | | | CDD18.3 | **49** | **0** | **0** | **0** | **0** | **1** | **10** | **19** | **13** | **6** | **0** | **0** | **0** | (11) |
| (12) | | | CDH23.3 | **374** | **0** | **0** | **0** | **2** | **16** | **87** | **157** | **83** | **29** | **1** | **0** | **0** | (12) |
| (13) | | | CDH26.7 | **71** | **0** | **0** | **0** | **0** | **1** | **17** | **30** | **21** | **2** | **0** | **0** | **0** | (13) |
| (14) | Wind | | WSAvg | **4.1** | **4.5** | **4.6** | **4.6** | **4.3** | **4.1** | **3.8** | **3.7** | **3.5** | **3.4** | **3.7** | **4.2** | **4.5** | (14) |
| (15) | Precipitation | | PrecAvg | **964** | **113** | **94** | **65** | **73** | **67** | **51** | **55** | **56** | **60** | **100** | **117** | **112** | (15) |
| (16) | | | PrecMax | **1197** | **246** | **225** | **205** | **186** | **165** | **123** | **149** | **113** | **156** | **150** | **241** | **218** | (16) |
| (17) | | | PrecMin | **714** | **26** | **19** | **8** | **3** | **6** | **4** | **13** | **6** | **3** | **18** | **20** | **25** | (17) |
| (18) | | | PrecStd | **132** | **58** | **52** | **43** | **45** | **35** | **30** | **30** | **30** | **40** | **36** | **55** | **53** | (18) |
| (19) | Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures | 0.4% | DB | | **13.1** | **14.7** | **19.8** | **23.0** | **25.2** | **29.0** | **29.7** | **29.6** | **26.3** | **20.6** | **16.6** | **14.4** | (19) |
| (20) | | | MCWB | | **12.0** | **11.3** | **11.7** | **15.2** | **17.8** | **21.3** | **20.0** | **21.2** | **19.0** | **17.1** | **14.7** | **12.8** | (20) |
| (21) | | 2% | DB | | **12.5** | **12.4** | **16.0** | **19.5** | **22.4** | **25.7** | **27.0** | **25.8** | **23.8** | **18.9** | **15.4** | **13.6** | (21) |
| (22) | | | MCWB | | **11.6** | **10.6** | **11.1** | **13.2** | **16.3** | **18.3** | **19.3** | **19.3** | **18.1** | **16.1** | **14.1** | **12.5** | (22) |
| (23) | | 5% | DB | | **12.0** | **11.8** | **13.7** | **17.1** | **20.0** | **23.6** | **24.9** | **23.1** | **21.9** | **17.7** | **14.5** | **13.0** | (23) |
| (24) | | | MCWB | | **11.1** | **10.4** | **10.4** | **12.1** | **14.9** | **17.6** | **18.5** | **18.0** | **17.1** | **15.6** | **13.3** | **12.0** | (24) |
| (25) | | 10% | DB | | **11.4** | **11.1** | **12.4** | **14.9** | **17.8** | **21.4** | **22.6** | **21.3** | **20.1** | **16.8** | **13.7** | **12.3** | (25) |
| (26) | | | MCWB | | **10.4** | **9.9** | **10.3** | **11.2** | **14.0** | **16.4** | **17.4** | **17.2** | **16.4** | **15.1** | **12.5** | **11.3** | (26) |
| (27) | Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures | 0.4% | WB | | **12.7** | **12.1** | **12.9** | **15.9** | **18.7** | **21.6** | **21.3** | **22.1** | **20.6** | **18.0** | **15.8** | **13.5** | (27) |
| (28) | | | MCDB | | **12.9** | **13.2** | **16.4** | **20.6** | **23.7** | **27.3** | **27.6** | **28.7** | **25.0** | **19.7** | **16.0** | **13.8** | (28) |
| (29) | | 2% | WB | | **12.0** | **11.5** | **12.0** | **14.1** | **16.8** | **19.5** | **20.0** | **19.9** | **19.0** | **17.0** | **14.4** | **12.9** | (29) |
| (30) | | | MCDB | | **12.4** | **12.1** | **13.8** | **18.2** | **21.1** | **24.4** | **25.8** | **23.9** | **22.0** | **18.0** | **15.0** | **13.4** | (30) |
| (31) | | 5% | WB | | **11.4** | **10.9** | **11.4** | **12.7** | **15.6** | **18.1** | **18.9** | **18.8** | **18.0** | **16.3** | **13.6** | **12.3** | (31) |
| (32) | | | MCDB | | **11.9** | **11.5** | **12.9** | **15.7** | **19.2** | **22.5** | **23.5** | **21.7** | **20.2** | **17.2** | **14.3** | **12.9** | (32) |
| (33) | | 10% | WB | | **10.7** | **10.1** | **10.8** | **11.8** | **14.4** | **16.9** | **18.0** | **18.0** | **17.2** | **15.5** | **12.7** | **11.6** | (33) |
| (34) | | | MCDB | | **11.3** | **11.0** | **12.1** | **14.1** | **17.1** | **20.3** | **21.4** | **20.2** | **19.2** | **16.5** | **13.6** | **12.3** | (34) |
| (35) | Mean Daily Temperature Range | | MDBR | | **5.3** | **5.9** | **6.8** | **7.8** | **8.0** | **8.3** | **8.0** | **8.0** | **8.2** | **6.6** | **5.9** | **5.2** | (35) |
| (36) | | 5% DB | MCDBR | | **4.9** | **5.9** | **9.1** | **11.3** | **11.3** | **12.4** | **12.6** | **10.9** | **10.6** | **7.4** | **6.2** | **4.3** | (36) |
| (37) | | | MCWBR | | **4.7** | **4.7** | **5.6** | **6.3** | **6.4** | **6.7** | **6.5** | **5.7** | **5.8** | **4.9** | **5.1** | **4.3** | (37) |
| (38) | | 5% WB | MCDBR | | **4.6** | **4.9** | **6.8** | **9.1** | **10.0** | **11.4** | **11.0** | **9.0** | **8.5** | **6.0** | **5.6** | **4.3** | (38) |
| (39) | | | MCWBR | | **4.8** | **4.6** | **4.8** | **5.6** | **6.1** | **6.6** | **6.1** | **5.3** | **5.3** | **4.7** | **5.3** | **4.5** | (39) |
| (40) | Clear-Sky Solar Irradiance | | taub | | **0.333** | **0.347** | **0.371** | **0.381** | **0.385** | **0.380** | **0.379** | **0.369** | **0.357** | **0.356** | **0.336** | **0.327** | (40) |
| (41) | | | taud | | **2.402** | **2.384** | **2.333** | **2.322** | **2.356** | **2.394** | **2.400** | **2.448** | **2.462** | **2.455** | **2.428** | **2.374** | (41) |
| (42) | | | Ebn at Noon | | **731** | **806** | **841** | **870** | **878** | **882** | **877** | **873** | **850** | **787** | **721** | **684** | (42) |
| (43) | | | Edh at Noon | | **69** | **87** | **106** | **118** | **118** | **115** | **113** | **103** | **92** | **80** | **67** | **63** | (43) |
| (44) | All-Sky Solar Radiation | | RadAvg | | **1.01** | **1.86** | **3.06** | **4.72** | **5.57** | **6.05** | **5.78** | **4.98** | **3.97** | **2.31** | **1.27** | **0.85** | (44) |
| (45) | | | RadStd | | **0.08** | **0.18** | **0.32** | **0.41** | **0.45** | **0.62** | **0.29** | **0.38** | **0.28** | **0.18** | **0.10** | **0.09** | (45) |

## Historical Trends

| | | DBAvg | Heating | | Cooling | | | Degree-Days | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | | 99% DB | 99% DP | 1% DB | 1% WB | 1% DP | HDD10.0 | HDD18.3 | CDD10.0 | CDD18.3 | |
| | | (d) | (e) | (f) | (g) | (h) | (i) | (j) | (k) | (l) | (m) | |
| (46) | Station Only | **N/A** | **N/A** | **N/A** | **N/A** | **N/A** | **N/A** | **N/A** | **N/A** | **N/A** | **N/A** | (46) |
| (47) | Regional (23 neighbors) | **N/A** | **N/A** | **N/A** | **N/A** | **N/A** | **N/A** | **N/A** | **N/A** | **N/A** | **N/A** | (47) |

Nomenclature: See separate page